MC: Monte-Carlo learning

for $S_1, A_1, R_1, S_2, A_2, R_2 \cdots$ under π

def: Return $G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^n R_T$ "Terminance"

def: Value Function: $V(s) = \mathbb{E}[G_t | S_t = s]$

Empirical Mean Return

1. Model-Free: no knowledge of MDP transition/reward
2. Must learning from complete episode
3. Goal: learn $V_\pi(s)$ (≠ not $V(s)$)

what we want: $V(s)$ converge to $V_\pi(s)$ as $N(s) \to \infty$.

Actually will. Since Law of Large Number

Every-visit MC: average return for every-time s is visited in an episode.
First-visit MC: average return for first time s is visited in an episode.

every time MC → "first time" for First time MC

For Every-time S is visited in an episode:

1. $N(s) = N(s) + 1$.
2. $S(s) = S(s) + G_t$. $\Rightarrow V(s) = S(s)/N(s)$

MC: Policy Evaluation

Input: a policy π to be evaluated
Initialize: in Practice: $V(s) = 0$.
$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in S$
$Returns(s) \leftarrow$ an empty list, for all $s \in S$

Loop forever (for each episode):
Generate an episode following π: $S_0, A_0, R_1, S_1, A_1, R_2, \ldots, S_{T-1}, A_{T-1}, R_T$
$G \leftarrow 0$
Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
$G \leftarrow \gamma G + R_{t+1}$
Unless S_t appears in $S_0, S_1, \ldots, S_{t-1}$:
Append G to $Returns(S_t)$
$V(S_t) \leftarrow \text{average}(Returns(S_t))$

$V(s) = S(s) / N(s)$

where $S(s) = G_{t_1} + G_{t_2} + \cdots + G_{t_n}$

Do it incrementally

$V(s) = V(s) + \frac{1}{N(s)}(G_t - V(s))$

$$V(s_t)_{org} = \frac{G_1 + G_2 + G_3}{3}.$$

$$V(s_t)_{update} = \frac{G_1 + G_2 + G_3 + G_4}{4} = \frac{G_1 + G_2 + G_3}{4} + \frac{G_4}{4}$$
$$= \frac{G_1 + G_2 + G_3}{3} \cdot (1 - \frac{1}{4}) + \frac{G_4}{4}$$
$$= V(s_t)_{org} + \frac{1}{4}(G_4 - V(s_t)_{org})$$

In nonstationary Problem (i.e. Policy, reward would change)

$V(s) = V(s) + \alpha(G_t - V(s))$

- MC works when the environment is unknown
- Working with a sampled episode has a huge advantage, even when one has complete knowledge of the environment's dynamics, for example, when transition probability is complex to compute
- Cost of estimating a single state's value is independent of the total number of states. We can sample an episode starting from the state of interest, then average returns.

if: Model unavailable.

We better estimate State-action Value (i.e. $Q(s,a)$)

Challenge: Some action-State pair may never be used.

Solution: Exploring Starts (try every pairs at each episode's Start)

TD: temporary Diffference Leanning

MC: Estimate using actual emperical rewards

TD: Using estimated value $V(s_{t+1})$ to estimate

⇒ "Bootstrapping"

TD target

$V(s_t) = V(s_t) + \alpha[R_t + \gamma V(s_{t+1}) - V(s_t)]$ ← TD error

Update a guess towards a guess

MC	TD
$V(s_t) = V(s_t) + \alpha(G_t - V(s_t))$	$V(s_t) = V(s_t) + \alpha(R_t + \gamma V(s_{t+1}) - V(s_t))$
Zero Bias & High Variance	Low Bias & Low Variance
Less sensitive to initial value	Sensitive to initial value
need complete Sequence	incomplete sequence is also ok
need episodic environment	can use in continuous env

TD target is much lower variance than the return:
- Return depends on **many** random actions, transitions, rewards
- TD target depends on **one** random action, transition, reward

more efficient

always Converge.

$V_\pi(s) = \mathbb{E}[G_t | S_t = s]$

MC: want $V(s_t) \to G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} \cdots + \gamma^n R_T$

$\therefore \mathbb{E}[V(s_t)] = \mathbb{E}[G_t | S_t = s] = V_\pi(s)$

Unbiased

Variance

MC: Sequence Size ↑. Variance ↑.

TD: only depend on R_t

TD: want $V(s_t) \to R_t + \gamma V(s_{t+1})$

(However: $G(s) = R_t + \gamma V_\pi(s_{t+1})$]

$\Rightarrow \mathbb{E}[V(s_t)] \neq \mathbb{E}[G(s_t)]$

$\Rightarrow \mathbb{E}[V(s_t)] \neq V_\pi(s_t)$

$\therefore$ Biased

Both MC & TD

finite experience, Solution: repeat the episode K times

TD exploits Markov Property while MC not.

Markov 属性是指：

> 当前状态已经包含了预测未来所需的所有信息，过去的历史不再重要。

形式化地说：

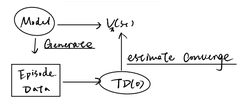
$$P(S_{t+1}, R_{t+1} \mid S_t, A_t) = P(S_{t+1}, R_{t+1} \mid S_0, A_0, \ldots, S_t, A_t)$$

We Assume: $V(s_{t+1})$ contains all info from future.

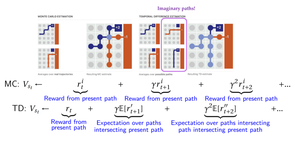
TD(0): $V(s_t) \leftarrow V(s_t) + \alpha(R_t + \gamma V(s_{t+1}) - V(s_t))$

Converges to a Solution of Max Likelihood Markov Model

Assume: There is a Markov Model



Certainty Equivalence estimate



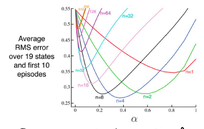
- Bootstrapping: update involves an estimate
  - MC does not bootstrap
  - DP bootstraps
  - TD bootstraps
- Sampling: update samples an expectation
  - MC samples
  - DP does not sample
  - TD sample

TD(λ)

"Look n steps into the future"

TD(0): no look at future

TD(1): look all future ⇒ MC. (n=∞)



Best way: Combining TD & MC

n-Step return

$G_t^{(n)} = R_t + \gamma R_{t+1} + \cdots + \gamma^{n-1} R_{t+n-1} + \gamma^n V(s_{t+n})$

λ-return

$G_t^\lambda = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$

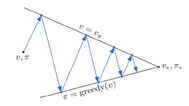
$\sum (1-\lambda)\lambda^{n-1} = 1$ ⇒ 全开就是1了

Forward-view TD(λ)

$V(s_t) \leftarrow V(s_t) + \alpha(G_t^\lambda - V(s_t))$ ⇒ Too many computation!

Model-Free Control

① On-policy MC Control



- Policy evaluation estimate V_π ⟶ Using MC, TD, DP, ⋯
  - Iterative policy evaluation
- Policy improvement generate $\pi' \geq \pi$
  - Greedy policy improvement

Evaluation: MC.

Improvement: Greedy: $\pi(s) = \arg\max_a [r(s,a) + \gamma \sum_{s'} p(s'|s,a) V(s')]$ require Model(p, r)

$\pi(s) = \arg\max_a Q(s,a)$. Model Free

- Greedified policy meets the conditions for policy improvement:
- For all $s \in S$:

$$Q_{\pi_k}(s, \pi_{k+1}(s)) = Q_{\pi_k}\left(s, \arg\max_a Q_{\pi_k}(s,a)\right)$$
$$\geq Q_{\pi_k}(s, \pi_k(s))$$

- And thus must be $\pi_{k+1} \geq \pi_k$

Two assumption to obtain Convergence of MC.

1. Exploring Start
2. Policy Evaluation with infinite number of episodes.

- The exploration-exploitation dilemma!
- If we always follow the **deterministic policy** to collect experience, we may never have the opportunity to see and evaluate (estimate Q) of alternative actions ...
- Solutions:
  1. **Exploring starts**: Every state-action pair has a non-zero probability of being the starting pair
  2. **Stochastic policies**: Give up on deterministic policies and only search over stochastic or soft polices (e.g., ϵ-greedy)
  3. **Off-policy learning**: Use a different policy to collect experience than the one you care to evaluate.

Solution 1: Exploring Start

Solution 2: ε-greedy

$\pi'(a_t|s_t) = \begin{cases} \frac{\epsilon}{|A|} + 1 - \epsilon, & \text{if } a^* = \arg\max_a Q_\pi(s_t, a) \\ \frac{\epsilon}{|A|}, & \text{otherwise} \end{cases}$

$V_{\pi'}(s) = Q_\pi(s, \pi'(s))$

$= \sum_{a \in A} \pi'(a|s) Q_\pi(s,a)$

$= \frac{\epsilon}{|A|} \sum_{a \in A} Q_\pi(s,a) + (1-\epsilon) \cdot \arg\max_a Q_\pi(s_t, a)$

$\geq \frac{\epsilon}{|A|} \sum_{a \in A} Q_\pi(s,a) + (1-\epsilon) \sum_{a \in A} \frac{\pi(a|s) - \frac{\epsilon}{|A|}}{1-\epsilon} Q_\pi(s,a)$

$= \frac{\epsilon}{|A|} \sum_{a \in A} Q_\pi(s,a) + \sum_{a \in A} \pi(a|s) Q_\pi(s,a) - \frac{\epsilon}{|A|} \sum_{a \in A} Q_\pi(s,a)$

$= \sum_{a \in A} \pi(a|s) Q_\pi(s,a) = V_\pi(s)$.

Actually: $\arg\max_a Q_\pi(s_t, a) \geq Q_\pi(s_t, a)$ for any a. (equals to 1)

$\therefore$ ε-greedy always improves policy

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
Initialize:
$\pi \leftarrow$ an arbitrary ε-soft policy
$Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in S$, $a \in A(s)$
$Returns(s,a) \leftarrow$ empty list, for all $s \in S$, $a \in A(s)$
Repeat forever (for each episode):
Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
$G \leftarrow 0$
Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
$G \leftarrow \gamma G + R_{t+1}$ ← MC
Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \ldots, S_{t-1}, A_{t-1}$:
Append G to $Returns(S_t, A_t)$
$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
$A^* \leftarrow \text{argmax}_a Q(S_t, a)$ (with ties broken arbitrarily) ← ε-greedy
For all $a \in A(S_t)$:
$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|A(S_t)| & \text{if } a = A^* \\ \varepsilon/|A(S_t)| & \text{if } a \neq A^* \end{cases}$

GLIE Theorem (Greedy in the Limit with Infinite Exploration)

Definition (GLIE)

- All state-action pairs are explored infinitely many times, $\lim_{k\to\infty} N_k(s,a) = \infty$ MC guaranteed this
- The policy converges on a greedy policy, $\lim_{k\to\infty} \pi_k(a|s) = \mathbb{1}(a = \arg\max_{a' \in A} Q_k(s,a'))$ Q-greedy $p(a^*|s) = \frac{\epsilon}{|A|} + 1 - \epsilon \Rightarrow$ if $\epsilon = \frac{1}{k}$, $\lim_{k\to\infty} p(a^*|s) = 1$

Theorem (GLIE Convergence Theorem)

GLIE model-free control converges to the optimal action-value functions, $Q_t \to Q_*$

- Sample k-th episode using π: $\{s_1, a_1, r_2, \ldots, s_T\} \sim \pi$
- For each state s_t and action a_t in the episode,

$$N(s_t, a_t) \leftarrow N(s_t, a_t) + 1$$
$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \frac{1}{N(s_t, a_t)}(G_t - Q(s_t, a_t))$$

- Improve policy based on new action-value function

$$\epsilon \leftarrow \frac{1}{k}, \quad \pi \leftarrow \epsilon\text{-greedy}(Q)$$

- This GLIE Monte-Carlo control converges to the optimal action-value function, $Q(s,a) \to Q_*(s,a)$

On-policy learning

"π" ⟶ Data → policy eva / policy imp. → Better "π"

Off-policy learning

another policy "μ" ⟶ Data ⟶ a good policy "π"

- On-policy Learning
  - "Learn on the job"
  - Learn about behavior policy π from experience sampled from π
  - On-policy methods attempt to evaluate or improve the policy that is used to make decisions.
- Off-policy Learning
  - "Look over someone's shoulder"
  - Learn about target policy π from experience sampled from μ
  - Off-policy methods evaluate or improve a policy different from that used to generate the data.
- Why is this important?
  - Learn from observing humans or other agents (e.g., from logged data)
  - Re-use experience from old policies (e.g., from past experience)
  - Learn about multiple policies while following one policy
  - Learn about greedy policy while following exploratory policy

$\mu(a|s)$ generate → action a, ⟶ $V_\pi(s)$, $Q_\pi(s,a)$ evaluate → $\pi(a|s)$.

How?

① Importance Sampling

$\mathbb{E}_{x\sim p}[f] = \int f(x) p(x) dx \approx \frac{1}{N} \sum_{n=1}^{N} f(x_n)$ "Sampling"

Law of Large Number

$\mathbb{E}_{x\sim p}[f] = \int f(x) p(x) dx$

$= \int f(x) \frac{p(x)}{q(x)} q(x) dx$.

$\therefore \mathbb{E}_{x\sim q}[f(x) \frac{p(x)}{q(x)}] \approx \frac{1}{N} \sum_{n=1}^{N} \frac{p(x_n)}{q(x_n)} f(x_n)$

$w_n = \frac{p(x_n)}{q(x_n)}$ are "importance weights"

$x \sim q$ sampling → $x_1, x_2, \cdots, x_n$ → $p(x_n), q(x_n)$ correct → $x \sim p$.

Summary:

- Estimate the expectation of a function

$$\mathbb{E}_{x\sim p}[f(x)] = \int f(x) p(x) dx \approx \frac{1}{n} \sum_i f(x_i)$$

- But sometimes it is difficult to sample x from $p(x)$, then we can sample x from another distribution $q(x)$, then correct the weight:

$$\mathbb{E}_{x\sim p}[f(x)] = \int f(x) p(x) dx = \int q(x) \frac{p(x)}{q(x)} f(x) dx \approx \frac{1}{n} \sum_i \frac{p(x_i)}{q(x_i)} f(x_i)$$

Importance Sampling for Off-policy RL

$\mathbb{E}_{\tau\sim\pi}[G(\tau)] = \int \pi(\tau) G(\tau) d\tau$ $\tau \sim \pi$ τ is a trajectory!

$= \int \mu(\tau) \frac{\pi(\tau)}{\mu(\tau)} G(\tau) d\tau$.

$\approx \frac{1}{N} \sum_{i=1}^{N} \frac{\pi(\tau_i)}{\mu(\tau_i)} G(\tau_i)$, $\tau_i \sim \mu$

- Given a starting state s_t, the probability of the subsequent state-action trajectory $a_t, s_{t+1}, a_{t+1}, \ldots, s_T$, occurring under any policy π

$$p(a_t, s_{t+1}, a_{t+1}, \ldots, s_T | s_t, a_{t:T-1} \sim \pi)$$
$$= \pi(a_t|s_t) p(s_{t+1}|s_t, a_t) \pi(a_{t+1}|s_{t+1}) \ldots p(s_T|s_{T-1}, a_{T-1})$$
$$= \prod_{k=t}^{T-1} \pi(a_k|s_k) p(s_{k+1}|s_k, a_k)$$

- Importance sampling ratio ρ

$$\frac{\pi(\tau)}{\mu(\tau)} = \rho_t^{T-1} = \frac{\prod_{k=t}^{T-1} \pi(a_k|s_k) p(s_{k+1}|s_k, a_k)}{\prod_{k=t}^{T-1} \mu(a_k|s_k) p(s_{k+1}|s_k, a_k)} = \prod_{k=t}^{T-1} \frac{\pi(a_k|s_k)}{\mu(a_k|s_k)}$$

$\mathbb{E}_{a\sim\mu}\left[\frac{\pi(a|s_k)}{\mu(a|s_k)}\right] = \sum_a \mu(a|s_k) \cdot \frac{\pi(a|s_k)}{\mu(a|s_k)} = \sum_a \pi(a|s_k) = 1$.

all ρ have expected value 1 (but since ρ may have value 500/1000/⋯ ρ can have high variance).

What we have → $\mathbb{E}(G_t | S_t = s) = V_\mu(s)$

What we want → $\mathbb{E}(\rho_t^{T-1} G_t | S_t = s) = V_\pi(s)$

Ordinary importance Sampling

$V(s) = \frac{\sum_{t\in\tau(s)} \rho_t^{T(t)} G_t}{|\tau(s)|}$.

Unbiased → w/ less # samples

$S_1, a_1, S_2, a_2 \cdots S_t, a_{t+1} \cdots S_{T-1}, a_T, S_T$

$V(s_t) = \frac{\sum_{t\in\tau(s)} \rho_t^{T(t)} G(t)}{|\tau(s)|}$.

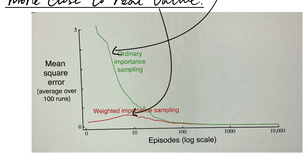
Weighted IS

$V(s) = \frac{\sum_{t\in\tau(s)} \rho_t^{T(t)} G(t)}{\sum_{t\in\tau(s)} \rho_t^{T(t)-1}}$ Biased → more close to real value.

$= \sum_{t\in\tau(s)} \frac{\rho_t^{T(t)}}{\sum \rho_t^{T(t)-1}} G(t)$

a "weight" $\in (0, 1]$.

may far from real value (High Var)



Ordinary IS may have infinite Variance

On-policy TD: low-variance, Online. Incomplete Seq.

① evaluation: SARSA (Update while interacting with env)

$Q(s,a) \leftarrow Q(s,a) + \alpha(r(s,a) + \gamma Q(s',a') - Q(s,a))$

S → A → R → S → A (SARSA)

② improvement: ε-greedy

Convergence of SARSA:

Conditions: ① GLIE of $\pi_t(a|s)$
② $\sum_{t=1}^{\infty} \alpha_t = \infty$, $\sum_{t=1}^{\infty} \alpha_t^2 < \infty$
where α is step size (learning rate)

n-Step SARSA

n-Step Q-return

$Q_t^{(n)} = R_t + \gamma R_{t+1} + \cdots + \gamma^{n-1} R_{t+n-1} + \gamma^n Q(S_{t+n}, A_{t+n})$.

n-Step SARSA

$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(Q_t^{(n)} - Q(s_t, a_t))$.

SARSA(λ) $Q_t^\lambda = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} Q_t^{(n)}$ ↑ alternate Q_t^λ

Off-policy TD Control

$V_\pi(s_t) \leftarrow V_\pi(s_t) + \alpha\left(\frac{\pi(a_t|s_t)}{b(a_t|s_t)}(R_t + \gamma V_\pi(s_{t+1})) - V_\pi(s_t)\right)$

Much lower Variance

Policies only need to be similar over a single step

MC: $\rho_t^{T-1} = \prod_{k=t}^{T-1} \frac{\pi(a_k|s_k)}{\mu(a_k|s_k)}$

TD: only $\frac{\pi(a_t|s_t)}{\mu(a_t|s_t)}$

$\mathbb{E}_\mu\left[\frac{\pi(a_t|s_t)}{b(a_t|s_t)}(R_t + \gamma V_\pi(s_{t+1})) - V_\pi(s_t) | S_t = s\right]$

$= \sum_a \mu(a|s) \left(\frac{\pi(a_t|s_t)}{\mu(a_t|s_t)} \mathbb{E}[R_t + \gamma V_\pi(s_{t+1}) | S_t = s, a_t = a]\right) - V_\pi(s)$ (r(s,a))

$= \sum_a \pi(a|s) \mathbb{E}[R_t + \gamma V_\pi(s_{t+1}) | S_t = s, a_t = a] - \sum_a \mu(a|s) V_\pi(s)$ (Σμ = 1)

$= \sum_a \pi(a|s) \mathbb{E}[R_t + \gamma V_\pi(s_{t+1}) | S_t = s, a_t = a] - V_\pi(s)$ (Σπ = 1)

$= \sum_a \pi(a|s) (\mathbb{E}[R_t + \gamma V_\pi(s_{t+1}) | S_t = s, a_t = a] - V_\pi(s))$

$= \mathbb{E}_\pi[R_t + \gamma V_\pi(s_{t+1}) - V_\pi(s) | S_t = s]$

Q-learning

$Q(s,a) \leftarrow Q(s,a) + \alpha[r(s,a) + \gamma \max_{a'} Q(s',a') - Q(s,a)]$.

$S \to a \to r \to s' \to a'$ Off-line

Q-learning: choose a' with $\max Q(s',a')$

SARSA Online

$Q(s,a) \leftarrow Q(s,a) + \alpha[r(s,a) + \gamma Q(s',a') - Q(s,a)]$

Why Q-learning do not need IS

Because Q-learning doesn't make expected value estimates over the policy distribution

Bellman Optimality Backup in Value Iteration

$Q(s,a) = r(s,a) + \gamma \sum_{s'\in S} p(s'|s,a) \max_{a'} Q(s',a')$

☆ Q-learning: a Sample-based Value iteration

Sample ⇒ $p(s'|s,a)$: over transition distribution not $\mu(a|s)$ ⇒ So no need of IS

Expected SARSA

$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[R_t + \gamma \mathbb{E}[Q(s_{t+1}, a_{t+1}) | s_{t+1}] - Q(s_t, a_t)]$

$\leftarrow Q(s_t, a_t) + \alpha[R_t + \gamma \sum \pi(a|s_{t+1}) Q(s_{t+1}, a) - Q(s_t, a_t)]$

perform better than SARSA (but Cost more)

Why? is it on-policy or off-policy?

Can be used both!

if π: Greedy. ⇒ General Q-learning

Q-Learning Variant

maximization bias

$\mathbb{E}[\max_i \mu_i] \geq \max_i \mathbb{E}(\mu_i)$

we use the same Q to both choose argmax and to evaluate itself

Double Q-learning

traing two action-value function Q_1 & Q_2.

$Q_1(s_t, a_t) \leftarrow Q_1(s_t, a_t) + \alpha(R_t + Q_2(s_{t+1}, \arg\max_a Q_1(s_{t+1}, a)) - Q_1(s_t, a_t))$.

$Q_2(s_t, a_t) \leftarrow Q_2(s_t, a_t) + \alpha(R_t + Q_1(s_{t+1}, \arg\max_a Q_2(s_{t+1}, a)) - Q_2(s_t, a_t))$.

training Q_1: using Q_1 to select greedy action
using Q_2's action-value function to update

Inference: ε-greedy using sum (Q_1, Q_2)

Double-Q estimation $\leq \max_i \mathbb{E}[X_i] \leq \mathbb{E}[\max_i X_i]$ ← Q-learning